

4.16 Do Now: Normal and Binomial Distribution Practice — Solutions

1. The heights, H cm, of sunflowers grown in a garden competition are normally distributed with a mean of 148 cm and a standard deviation of 11 cm.
 - (a) Find $P(H < 140)$. [2]
 - (b) Find $P(135 < H < 160)$. [2]
 - (c) The tallest 20% of sunflowers receive a prize. Find the minimum height, in cm, required to receive a prize. [2]
 - (d) Given that a sunflower does not receive a prize, find the probability that it is taller than 150 cm. [2]

Solutions

(a) $\text{normCdf}(0, 140, 148, 11) = 0.23352 \dots$

$\approx \mathbf{0.234}$

(b) $\text{normCdf}(135, 160, 148, 11) = 0.74370 \dots \approx \mathbf{0.744}$

(c) $\text{invNorm}(0.80, 148, 11) = 157.258 \dots \approx \mathbf{157}$ cm

(d) $\text{normCdf}(150, 157.258 \dots, 148, 11) = 0.22786 \dots$

$$\frac{0.22786 \dots}{0.80} = 0.28482 \dots \approx \mathbf{0.285}$$

2. A city bus arrives on time at a particular stop with probability 0.65. Over a period of one week, the bus makes 20 scheduled arrivals. The number of times the bus arrives on time is denoted by X .
- (a) Find $P(X = 13)$. [2]
 - (b) Find $P(X \geq 15)$. [2]
 - (c) Find $E(X)$ and $\text{Var}(X)$. [2]
 - (d) Find the greatest integer value of n such that $P(X \geq n) > 0.95$. [3]
 - (e) Explain in the context of the problem what the value of n found in part (d) represents. [1]

Solutions

(a) $\text{binomPdf}(20, 0.65, 13) = 0.18440 \dots \approx \mathbf{0.184}$

(b) $1 - \text{binomCdf}(20, 0.65, 14) = 0.24539 \dots \approx \mathbf{0.245}$

(c) $E(X) = np = 20 \times 0.65 = \mathbf{13}$ $\text{Var}(X) = np(1 - p) = 20 \times 0.65 \times 0.35 = \mathbf{4.55}$

(d) $P(X \geq 9) = 1 - \text{binomCdf}(20, 0.65, 8) = 0.98042 \dots > 0.95 \checkmark$

$P(X \geq 10) = 1 - \text{binomCdf}(20, 0.65, 9) = 0.94683 \dots < 0.95$ so $n = \mathbf{9}$

(e) There is a greater than 95% probability that the bus arrives on time at least 9 times in 20 scheduled arrivals.